Bay Study Example Calculations

Preliminary scheme-comparison calcs — bay_studies.sqlite

October 27, 2026

Scope

Each calc below walks one representative bay from the bay_studies.sqlite database by hand, using the same assumptions as the design engine, so a reviewer can trace every number in that row. Each calc names its source table and row so it can be pulled with a query.

These are preliminary bay-study calcs for scheme comparison, not construction documents. Engine assumptions (loads, materials, code basis) are listed in the database's assumptions table; only the ones that drive each calc are repeated here. Embodied carbon is A1–A3 for the floor framing only, using the factors in the carbon_factors table.

Calc	System	Bay	Result	kg CO_2e/m^2
1	Composite steel floor	30 × 30 ft	W14x26 @ 10 ft / W21x48	90.7
2	Steel roof, 1.5B deck	30 × 30 ft	W12x19 or 22KCS2 @ 7.5 ft / W18x35	44.7 / 38.9
3	Mass timber floor	30 × 24 ft	CLT on 6-3/4 × 31-1/2 / 8-3/4 × 36	55.1
4	RC flat plate	24 × 24 ft	9 in plate, 20 in columns	94.2
5	PT flat plate	28 × 28 ft	7-1/2 in plate + stud rails	77.5
6	One-way slab on PT beams	48 × 16 ft	7-1/2 in slab, 14 × 24 PT beam	95.2
7	Pan joists on band beams	36 × 28 ft	14+4.5 @ 36 in, 36 × 18-1/2 band	91.9

Calc 1 — Composite steel floor (30 × 30 ft bay)

Result: W14x26 beams at 10 ft o.c. with 12 studs and 1-1/4” camber, on W21x48 girders with 18 studs and 1” camber; 4.20 psf steel, 27.1” structural depth, walking ap/g = 0.38% (limit 0.50%). Source: bay_designs row 829, STL_COMP, criterion “Walking b=0.03” (strength LL = 50 psf governs).

Basis. AISC 360-16 LRFD, AISC Design Guide 11 (2nd ed.). 2VLI deck + 4-1/2” NW topping (6-1/2” total), f’c = 4 ksi, wc = 145 pcf, A992 steel. Slab + deck 69 psf, SDL 15 psf, LL 50 psf, construction LL 20 psf, unshored.

Beam W14x26 (Lj = 30 ft, S = 10 ft)

W14x26: A = 7.69 in², d = 13.9 in, Ix = 245 in⁴, Zx = 40.2 in³.

Check	Demand	Capacity / limit	Ratio
Construction, 1.2(0.716) + 1.6(0.20) klf	Mu = 132.7 k-ft	φMp = 150.8 k-ft	0.88
Composite, wu = 1.2(0.866) + 1.6(0.50) = 1.84 klf	Mu = 206.9 k-ft	φMn = 233.0 k-ft	0.89
Live-load deflection	0.58 in	L/360 = 1.00 in	0.58
SDL + LL deflection	0.76 in	L/240 = 1.50 in	0.50

Wet-concrete deflection on the bare beam, w = 69(10) + 26 = 716 plf:

$$\Delta_{wet} = \frac{5wL^4}{384EI_x} = \frac{5(716/12)(360)^4}{384(29{,}000)(245)} = 1.84 \text{ in } \Rightarrow \text{ camber} = 0.8(1.84) \rightarrow 1\tfrac{1}{4} \text{ in}$$

Composite strength: b_eff = min(L/4, S) = 90 in. Studs are 3/4” in the weak rib position, Qn = Rg·Rp·Asc·Fu = 1.0(0.6)(0.442)(65) = 17.2 k. With 6 studs per half span, ΣQn = 103.4 k (27% composite, above the 25% minimum). Concrete block a = 103.4 / (0.85·4·90) = 0.34 in, so Y2 = 6.5 − 0.17 = 6.33 in. The steel section carries the remaining tension with the plastic P-M approximation Ms = 1.18Mp(1 − C/Py) = 1,734 k-in:

$$\phi M_n = 0.9\,[\,\Sigma Q_n(d/2 + Y_2) + M_s\,] = 0.9\,[\,103.4(6.95 + 6.33) + 1{,}734\,]/12 = 233 \text{ k-ft}$$

Deflection uses 0.75·Ieff (AISC Commentary I3.2): n = 8.30, Itr = 1,161 in⁴, Ieff from the √(ΣQn/Cf) interpolation, 0.75·Ieff = 540 in⁴.

Girder W21x48 (Lg = 30 ft, beams at third points)

W21x48: d = 20.6 in, Ix = 959 in⁴, Zx = 107 in³. Each point load carries 10 × 30 = 300 ft² (beams from both sides): PD = 21.5 k, PSD = 4.5 k, PL = 15.0 k.

Check	Demand	Capacity / limit	Ratio
Construction	Mu = 360 k-ft	φMp = 401 k-ft	0.90
Composite, Pu = 1.2(26.0) + 1.6(15.0) = 55.2 k	Mu = 558 k-ft	φMn = 583 k-ft	0.96
Live-load deflection	0.52 in	L/360 = 1.00 in	0.52

Girder studs: deck ribs parallel, Rp = 0.75, Qn = 21.5 k; 9 per half span gives ΣQn = 194 k. Wet deflection 1.31 in → 1” camber.

Walking vibration (DG11 Ch. 4, β = 0.03)

Vibration loads are slab 69 + SDL 4 + live 11 psf + steel self-weight. Dynamic modulus 1.35Ec gives n = 6.15.

Mode	I (in⁴)	w (plf)	Δ (in)	Panel width (ft)	W (kips)
Beam	1,280	866	0.425	Bj = 40.7	158.5 (×1.5 continuity)
Girder	3,792	2,646	0.438	Bg = 54.2	143.3

$$f_n = 0.18\sqrt{\frac{386}{0.425 + 0.438}} = 3.81 \text{ Hz}, \qquad \frac{a_p}{g} = \frac{65\, e^{-0.35(3.81)}}{0.03\,(152{,}000)} = 0.38\% < 0.50\%$$

Because Lg = 30 ft < Bj = 40.7 ft, the girder deflection is reduced by 30/40.7 when weighting the panel weights (W = 152 k), but not in fn (DG11 Eq. 4-6).

Carbon and schedule

Item	Quantity	Factor	kg CO₂e/m²
Framing steel	4.20 psf = 20.5 kg/m²	1,220 kg/t	25.0
Composite deck	2.0 psf = 9.8 kg/m²	2,320 kg/t	22.7
Concrete 4 ksi	5.5 in equivalent = 0.140 m³/m²	308 kg/m³	43.0
Total			**90.7**

Schedule per 20,000 sf floor: pieces = 20,000 × (1/300 + 1/900) = 89, at 35 picks/day = 2.5 days, plus deck at 8,000 sf/day = 2.5 days, so about 5.0 days. Lead time 12 weeks.

Calc 2 — Steel roof (30 × 30 ft bay, 1.5B deck, 30 psf snow)

Result: 1.5B 20 ga deck at 7’-6” spans, framed either with W12x19 beams (3.70 psf steel, 44.7 kg CO_2e/m^2) or 22KCS2 joists (2.50 psf, 38.9 kg CO_2e/m^2), both on W18x35 girders. Source: bay_designs rows 3067 (STL_ROOF_W) and 3068 (STL_ROOF_J).

Basis. AISC 360-16 LRFD for W shapes; SJI KCS table (ASD values × 1.5 for LRFD); Vulcraft Grade 50 deck properties. Roof SDL 15 psf + deck 2.0 psf, snow 30 psf. Top flanges braced by deck. Limits: LL ≤ L/240, total ≤ L/180.

Deck spacing

1.5B 20 ga (Id = 0.197 in^4/ft, φMn− = 859 lb-ft/ft) has a computed SDI 3-span construction limit of 7’-8”, so the 30 ft girder is split into 4 bays at 7’-6”. In service, wu = 1.2(2 + 15) + 1.6(30) = 68.4 psf gives $0.1 \cdot wu \cdot S^2$ = 385 lb-ft/ft < 859, and LL deflection 0.19 in < S/240 = 0.38 in.

Option A — W12x19 beams

Check	Demand	Capacity / limit	Ratio
Flexure, wu = 1.2(0.147) + 1.6(0.225) = 0.536 klf	Mu = 60.3 k-ft	φMp = 92.6 k-ft	0.65
Snow deflection	1.09 in	L/240 = 1.50 in	0.73
Total deflection	1.80 in	L/180 = 2.00 in	0.90

Deflection governs: a W12x16 (Ix = 103 in^4) reaches about 2.27 in total, over L/180.

Option B — 22KCS2 joists

22KCS2: M = 488 k-in and V = 5,900 lb (ASD), 10 plf, Igross = 194 in^4.

Check	Demand	Capacity / limit	Ratio
Moment, wu = 1.2(137.5) + 1.6(225) = 525 plf	Mu = 709 k-in	1.5 × 488 = 732 k-in	0.97
End shear	Vu = 7,875 lb	1.5 × 5,900 = 8,850 lb	0.89
Snow deflection, I = 194/1.15 = 169 in^4	0.84 in	1.50 in	0.56
Total deflection	1.35 in	2.00 in	0.68

Girder W18x35 (3 point loads at 7.5 ft)

Each point carries 7.5 × 30 = 225 ft^2. With W beams: PD = 4.40 k, PL = 6.75 k, Pu = 16.1 k.

$$M_u = 1.5P_u(15) - P_u(7.5) + \frac{1.2w_gL^2}{8} = 241.1 + 4.7 = 245.8 \text{ k-ft} \ \leq \ \phi M_p = 249.4 \text{ k-ft} \ (0.99)$$

Snow deflection 1.05 in (L/240 = 1.50 in); total 1.78 in (L/180 = 2.00 in). With joists the point loads drop slightly (Pu = 15.8 k, Mu = 241 k-ft); the same girder is lightest.

Ponding (AISC Appendix 2 simplified): Cp + 0.9Cs = 0.29 > 0.25 with the W beams, so this bay relies on roof slope ≥ 1/4 in/ft with drains. The database reports this flag but does not size for it.

Depth, carbon and schedule

Item	W beams	KCS joists
Structural depth	max(12.2, 17.7) + 1.5 = 19.2 in	max(22, 2.5 + 17.7) + 1.5 = 23.5 in
Beam / joist steel	2.53 psf × 1,220 kg/t	1.33 psf × 1,430 kg/t
Girder steel	1.17 psf × 1,220 kg/t	1.17 psf × 1,220 kg/t
Steel carbon	22.0 kg/m²	16.3 kg/m²
Deck, 2.0 psf × 2,320 kg/t	22.7 kg/m²	22.7 kg/m²
Total carbon	**44.7 kg/m²**	**38.9 kg/m²**
Pieces per 20,000 sf	111 at 35/day = 3.2 days	111 at 50/day = 2.2 days
Roof cycle incl. deck (2.5 days)	5.7 days	4.7 days

Calc 3 — Mass timber floor (30 ft beams × 24 ft girders, office, 1-hr fire)

Result: 5-ply E1 CLT spanning 12 ft to 6-3/4 × 31-1/2 glulam beams, on 8-3/4 × 36 glulam girders; 2.35 bf/sf, 45.9” structural depth, 55.1 kg CO_2e/m^2. Source: bay_designs row 6516, MT_FLOOR, “Office + 1-hr fire”.

Basis. NDS 2018 ASD. CLT: PRG 320 E1 5-ply 6-7/8”, (FbS)eff = 10,400 lb-ft/ft, (EI)eff = 440×10^6 lb-in^2/ft, (GA)eff = 0.92×10^6 lb/ft, Vs = 2,480 lb/ft. Glulam 24F-V4 DF: Fb = 2,400 psi, Fv = 265 psi, E = 1.8×10^6 psi. Dead load = CLT 20.6 + 3” NW topping 37.5 + SDL 10 = 68.1 psf; LL = 50 + 15 partitions = 65 psf; CD = 1.0. Creep factor Kcr = 2.0 (CLT), 1.5 (glulam).

CLT panel (single span, S = 12 ft)

Check	Demand	Capacity / limit	Ratio
Moment, w = 133.1 psf	2,396 lb-ft/ft	10,400	0.23
Shear	799 lb/ft	2,480	0.32
Live-load deflection (bending + shear)	0.08 in	L/360 = 0.40 in	0.21
Long-term, 2.0ΔD + ΔL	0.26 in	L/240 = 0.60 in	0.43

The CLT Handbook bare-panel vibration limit is 17’-10”, so 12 ft is well inside it. The 12 ft spacing comes from splitting the 24 ft girder into two bays.

Beam 6-3/4 × 31-1/2 (L = 30 ft)

Section: A = 212.6 in^2, S = 1,116 in^3, I = 17,581 in^4, self-weight 52 plf. Loads: wD = 68.1(12) + 52 = 869 plf, wL = 65(12) = 780 plf. Braced by the CLT, so CL = 1.0; volume factor CV = 0.852.

Check	Demand	Capacity / limit	Ratio
Bending	fb = 1,994 psi	Fb′ = 2,400 × 0.852 = 2,046 psi	0.97
Shear (at full reaction)	fv = 175 psi	265 psi	0.66
Live-load deflection	0.45 in	1.00 in	0.45
Long-term, 1.5ΔD + ΔL	1.20 in	1.50 in	0.80
1-hr fire, 3 sides, char 1.8 in → 3.15 × 29.7	fb = 4,808 psi	2.85 Fb CV = 6,329 psi	0.76

Girder 8-3/4 × 36 (Lg = 24 ft, one point load at midspan)

Section: S = 1,890 in^3, I = 34,020 in^4, 77 plf. Point load from beams both sides: PD = 26.1 k, PL = 23.4 k. Laterally braced only at the beam, so lu = 12 ft gives CL = 0.980; CV = 0.838 governs.

$$f_b = \frac{M}{S} = \frac{302{,}365(12)}{1{,}890} = 1{,}920 \text{ psi} \;\le\; F_b' = 2{,}400 \times \min(C_L, C_V) = 2{,}011 \text{ psi} \; (0.95)$$

Shear fv = 122 psi (0.46). Live-load deflection 0.19 in (L/360 = 0.80 in); long-term 0.52 in (L/240 = 1.20 in). The 1-hr fire check on the charred 5.15 × 34.2 section passes, which is why this girder is 8-3/4 wide rather than the 6-3/4 × 40-1/2 that strength alone picks.

Walking vibration (information only for this column)

This office/fire design gives fn = 3.81 Hz and ap/g = 0.64% at β = 0.03, over the 0.5% limit. Use the database’s Walking β=0.03 column for this bay when vibration matters; it deepens the beams.

Depth, carbon and schedule

Structural depth = max(31.5, 36) + 6.875 + 3 = 45.9 in.

Item	Quantity	Factor	kg CO_2e/m^2
CLT	6.875 in = 0.175 m^3/m^2	134 kg/m^3	23.4
Glulam	2.35 bf/sf = 0.060 m^3/m^2	137 kg/m^3	8.2
Topping 4 ksi	3 in = 0.076 m^3/m^2	308 kg/m^3	23.5
Total (biogenic carbon excluded)			**55.1**

Schedule per 20,000 sf floor: 83 glulam pieces at 25/day = 3.3 days, plus CLT at 7,500 sf/day = 2.7 days, so 6.0 days; topping is poured off the critical path. Lead time 16 weeks.

Calc 4 — Mild-reinforced two-way flat plate (24 × 24 ft, residential)

Result: 9” flat plate on 20” square columns, no shear reinforcement; 2.59 psf rebar, 94.2 kg CO_2e/m^2. ACI minimum thickness governs, with edge punching close behind at 0.96. Source: bay_designs row 7373, CONC_RC_FP, “Residential 40 psf”.

Basis. ACI 318-19, Direct Design Method. f’c = 5 ksi NW (150 pcf), Grade 60, 3/4” cover, #5 bars. SDL 15 psf, LL 40 psf. Slab edge flush with the exterior column face, no edge beam.

Thickness

Exterior panel without edge beams (Table 8.3.1.1): ln = 24 − 20/12 = 22.33 ft, so h ≥ 268/30 = 8.93 in → 9.0 in. Self-weight 112.5 psf; wu = 1.2(112.5 + 15) + 1.6(40) = 217 psf.

$$M_o = \frac{w_u \ell_2 \ell_n^2}{8} = \frac{217(24)(22.33)^2}{8} = 324.7 \text{ k-ft}$$

Punching shear (d = 9 − 1.375 = 7.63 in)

Column	Vu (k)	Moment to shear	vu (psi)	φvc (psi)	Ratio
Interior, b0 = 110.5 in	123.8	γv = 0.40 × Msc 26.8 k-ft (DDM Eq. 8.10.7.3)	163	212	0.77
Edge, 3-sided, b1 = 23.8, b2 = 27.6 in	65.8	γv = 0.38 × 0.3Mo = 97.4 k-ft	204	212	0.96

φvc = 0.75 × 4 × λs × √5000 with λs = 1.0 (d < 10 in). An 8-1/2” plate fails the edge column (1.03), so 9” is the thinnest that works both ways. With headed stud rails the same 9” plate drops to 0.48 at the edge (row 7374).

Flexure (end span, column strip = 12 ft)

Location	DDM coefficient	Column strip share	Mu, column strip (k-ft)	As required (in^2)
Exterior negative	0.26	100%	84.4	2.40
Positive	0.52	60%	101.3	2.89
First interior negative	0.70	75%	170.5	4.92 (0.41 in^2/ft)

Middle strip: interior negative 56.8 k-ft (1.61 in^2) and positive 67.5 k-ft (1.91 in^2); minimum 0.0018bh = 2.33 in^2 per 12 ft strip governs at the middle strip.

Rebar quantity is averaged over end and interior spans in both directions, with column-strip top bars 0.33L and middle-strip top bars 0.22L at each support, and bottom bars 1.1L (laps included). That gives 1.27 + 1.32 = 2.59 psf, about 6% above the CRSI Handbook tabulation for the same span (see the validation tables).

Carbon and schedule

Item	Quantity	Factor	kg CO_2e/m^2
Concrete 5 ksi	9 in = 0.229 m^3/m^2	365 kg/m^3	83.4
Rebar	2.59 psf = 12.6 kg/m^2	854 kg/t	10.8
Total			**94.2**

Schedule: typical RC flat plate cycle 6 days per 20,000 sf floor, adjusted +0.5% for rebar density (2.59 vs. 2.5 psf reference) = 6.0 days. Lead time 3 weeks.

Calc 5 — Post-tensioned two-way flat plate (28 × 28 ft, residential, stud rails)

Result: 7-1/2” PT flat plate on 24” columns with headed stud rails; about 21 strands per 28 ft bay each way (P/A = 223 psi), 0.82 psf strand + 0.48 psf mild steel, 77.5 kg CO_2e/m^2. Span/depth governs. Source: bay_designs row 7694, CONC_PT_FP, “Residential 40 psf + stud rails”.

Basis. ACI 318-19 Ch. 8 and 24, load balancing. f’c = 5 ksi NW, 1/2” unbonded 270 ksi strand at fse = 175 ksi (26.8 k each), Grade 60 mild steel. SDL 15 psf, LL 40 psf. Without studs this bay needs an 11” plate (row 7693).

Thickness and balancing force

h = L/45 = 336/45 = 7.47 → 7.5 in (7” practical minimum). Self-weight 93.8 psf; wu = 1.2(108.8) + 1.6(40) = 194.5 psf.

End-span drape with the anchor at mid-depth and 1” cgs top and bottom: a = 0.75h − 1.5 = 4.13 in. Balance 75% of self-weight, wbal = 70.3 psf:

$$F = \frac{w_{bal} L^2}{8a} = \frac{70.3(28)^2}{8(4.13/12)} = 20.0 \text{ k/ft} \;\Rightarrow\; P/A = \frac{20{,}045}{12(7.5)} = 223 \text{ psi}$$

That is 0.75 strands per ft, about 21 strands per 28 ft bay in each direction (banded one way and distributed the other in practice; the dataset treats both alike).

Service stresses (full-width design strip, per ft)

Net service load = 93.8 + 15 + 40 − 70.3 = 78.4 psf; ln = 26 ft; Mo,net = 78.4(26)²/8 per ft; S = 112.5 in^3/ft.

Location	M (k-in/ft)	Tension, −P/A + M/S (psi)	Limit 6√f’c (psi)	Compression (psi)
First interior support, 0.70Mo	55.7	272	424	718 < 0.60f’c = 3,000
Positive, 0.52Mo	41.4	145	424	590

Both stay in Class U, so no force increase was needed.

Strength

Aps = 0.115 in^2/ft, dp = 6.5 in. Unbonded strand with L/h > 35: fps = fse + 10,000 + f’c·b·dp/(300Aps) = 196 ksi. Factored moments use the DDM end-span coefficients on wu (secondary moments ignored).

Location	Mu (k-in/ft)	φMn from strand (k-in/ft)	Mild steel added
First interior negative	138.1	127.1	Deficit 11 k-in/ft, plus ACI 8.6.2.3 minimum 0.00075·Acf over columns
Positive	102.6	127.1	Tension 145 psi > 2√f’c = 141 psi, so Nc/(0.5fy) minimum bonded steel

Mild steel totals 0.48 psf for both directions; integrity, edge and trim bars are not included and typically add 0.3–0.7 psf.

Punching shear with stud rails (d = 6.0 in)

Interior: b0 = 4(24 + 6) = 120 in, Vu = 151 k, unbalanced moment from Eq. 8.10.7.3 gives vu = 238 psi. With studs the limit is φ8√f’c = 424 psi (0.56). Without studs the PT capacity βp√f’c + 0.3fpc = 3.5(70, √f’c capped) + 0.3(223) = 312 psi, φvc = 234 psi, ratio 1.02, which is why studs are needed at 7.5”. Edge column with studs: 0.69.

Carbon and schedule

Item	Quantity	Factor	kg CO_2e/m^2
Concrete 5 ksi	7.5 in = 0.191 m^3/m^2	365 kg/m^3	69.5
PT strand (2 ways, 0.52 plf, +5%)	0.82 psf = 4.0 kg/m^2	1,500 kg/t (estimate)	6.0
Mild rebar	0.48 psf = 2.3 kg/m^2	854 kg/t	2.0
Total			**77.5**

Schedule: PT flat plate cycle 5 days per 20,000 sf, adjusted −6% for the light mild steel (0.48 vs. 0.8 psf reference) = 4.7 days. Lead time 4 weeks.

Calc 6 — One-way slab on PT beams (48 ft beams at 16 ft, office)

Result: 7-1/2” mild one-way slab spanning 16 ft to 14 × 24 PT beams with 17 strands (P/A = 347 psi); 2.47 psf mild steel, 0.58 psf strand, 8.7” equivalent concrete, 95.2 kg CO_2e/m^2. Source: bay_designs row 8127, CONC_OW_PTB, “Office 50+15 psf (Min depth)”.

Basis. ACI 318-19. f’c = 5 ksi NW, Grade 60, 1/2” unbonded strand at 26.8 k each. SDL 15 psf, LL 65 psf. 18” columns assumed for clear spans. Beam treated as an end span of a continuous run.

One-way slab (S = 16 ft, ln = 14.8 ft)

h ≥ ln/24 = 7.42 → 7.5 in (one end continuous, Table 7.3.1.1). wu = 1.2(93.8 + 15) + 1.6(65) = 234.5 psf, d = 6.5 in.

Location	Coefficient	Mu (k-in/ft)	As req’d (in^2/ft)	As used
End-span positive	1/14	44.2	0.127	0.162 (0.0018bh minimum)
First interior negative	1/10	61.9	0.179	0.179

Slab steel including distribution bars = 1.60 psf.

PT beam 14 × 24 (L = 48 ft)

T-section: flange b = 14 + 2·min(8 × 7.5, (192 − 14)/2, 46.5×12/8) = 134 in; A = 1,236 in^2, yb = 18.0 in, I = 37,000 in^4, Sb = 2,055 in^3, St = 6,174 in^3. L/h = 24 (the practical limit used).

Loads per beam: D = (93.8 + 15)(16) + stem 241 = 1,981 plf; L = 65(16) = 1,040 plf; wu = 4,041 plf.

Drape with the anchor at the centroid and 3” cgs at the high and low points: a = [(18.0 − 3) + (24 − 6)]/2 = 16.5 in. Balancing 75% of D needs F = 311 k (252 psi), but positive-moment tension on the small bottom section modulus then exceeds 7.5√f’c, so the engine raises F to 429 k (347 psi), provided by 17 strands (455 k). That balances about 2,050 plf, slightly more than the dead load:

$$f_{t,pos} = -\frac{P}{A} + \frac{w_{net}\,\ell_n^2/14}{S_b} = -347 + \frac{972.5(46.5)^2(12)/14}{2{,}055} = 530 \text{ psi} \le 7.5\sqrt{f'_c} = 530 \text{ psi}$$

Negative-moment tension is 62 psi; compression peaks at 1,575 psi (≤ 0.60f’c).

Check	Demand	Capacity	Result
Negative, ln^2/10, stem in compression	Mu = 874 k-ft	φMn,strand = 626 k-ft	Mild top bars for the 248 k-ft deficit
Positive, ln^2/14, flange in compression	Mu = 624 k-ft	φMn,strand = 765 k-ft	0.004·Act minimum bottom bars only
Shear at first interior, 1.15wu·ln/2	Vu = 108 k	φVc = 2√f’c·bw·d = 31.2 k	Stirrups: Vs = 102 k → about #4 two-leg at 5 in near supports
Live-load deflection, gross I	0.35 in	L/360 = 1.60 in	0.22

fps for unbonded strand at L/h ≤ 35 is fse + 10,000 + f’c·b·dp/(100Aps), capped at fse + 60,000 psi. Beam mild steel (bars + stirrups) = 0.87 psf of floor.

The shear and the >100% balance are both signs this beam is at its practical limit; the deeper “Min concrete” option for this bay in the database is the easier one to build.

Carbon and schedule

Item	Quantity	Factor	kg CO_2e/m^2
Concrete 5 ksi	7.5 slab + 1.2 stem = 8.7 in = 0.221 m^3/m^2	365 kg/m^3	80.7
Mild rebar	2.47 psf = 12.1 kg/m^2	854 kg/t	10.3
PT strand	0.58 psf = 2.8 kg/m^2	1,500 kg/t (estimate)	4.3
Total			**95.2**

Schedule: one-way slab + PT beam cycle 7 days per 20,000 sf, +3.5% for rebar density (2.47 vs. 2.0 psf reference) = 7.2 days. Lead time 4 weeks.

Calc 7 — Pan joists on band beams (36 ft joists, 28 ft beams, office)

Result: 30” pans + 6” ribs at 36” (14” pans + 4-1/2” slab = 18-1/2” deep) spanning to 36 × 18-1/2 flush band beams; square-end pans, 3.77 psf rebar, 8.22” equivalent concrete, 91.9 kg CO_2e/m^2. Source: bay_designs row 8606, CONC_PAN, “Office 50+15 psf (30” pans)“.

Basis. ACI 318-19 (joist construction, Sec. 9.8), CRSI standard forms. f’c = 5 ksi NW, Grade 60. SDL 15 psf, LL 65 psf. 4-1/2” top slab for 2-hr fire. Joist clear span = 36 − 3 = 33 ft (3 ft beam width assumed); beam clear span 26 ft.

Joist section

Rib 6” at the bottom, tapered 1:12 per side, so average width = 6 + 14/12 = 7.17 in. Rib depth 14 ≤ 3.5 × 6 = 21 in (ACI 9.8.1.2).

$$t_{eq} = 4.5 + \frac{7.17 \times 14}{36} = 7.29 \text{ in } \Rightarrow \; w_{sw} = 91.1 \text{ psf}$$

Per joist (3 ft): wD = (91.1 + 15)(3) = 318 plf, wL = 65(3) = 195 plf, wu = 1.2(318) + 1.6(195) = 694 plf.

Check	Demand	Capacity / limit	Result
Positive, $ln^2/14$, T-section b = 36 in, d = 17.4 in	Mu = 54.0 k-ft	As = 0.70 in^2 (bottom)	Tension-controlled
Negative, $ln^2/10$, rib bottom 6 in, d = 17.4 in	Mu = 75.6 k-ft	As = 1.04 in^2 (top)	c/d = 0.17 < 0.375
Shear at d, 1.15wu·ln/2 − wu·d	Vu = 12.2 k	φ(1.1)(2√f’c)(7.17)(17.4) = 14.6 k	0.83, no tapered ends
Live-load deflection (Ie, Bischoff)	0.52 in	ln/360 = 1.10 in	0.47
Long-term, 2.0ΔD + ΔL	1.37 in	ln/240 = 1.65 in	0.83

Deflection is calculated rather than taken from ACI Table 7.3.1.1 (ln/18.5 would need 21.4 in). Gross I = 7,214 in^4 per joist, Mcr = 25.1 k-ft, service Ma = 39.9 k-ft, so the section is cracked: Icr = 1,279 in^4 and Ie = 1,495 in^4 under full service load. The propped-span coefficient $wL^4/185EI$ is used for the end span.

Band beam 36 × 18-1/2 (L = 28 ft, flush with joists)

Tributary width = joist span 36 ft. Extra weight of the solid band over the joist floor it replaces = 420 plf.

Check	Demand	Capacity	Result
Loads	wD = 4,240 plf, wL = 2,340 plf	wu = 8,832 plf	
Negative, $ln^2/10$, d = 16 in	Mu = 597 k-ft	As = 9.37 in^2	Tension-controlled
Positive, $ln^2/14$, T-section	Mu = 426 k-ft		
Shear at d	Vu = 120 k	φVc = 61.1 k	Stirrups for Vs = 79 k (4-leg ties in the wide band)

The band keeps a flat soffit and the same 18-1/2” depth; a 14” joist was chosen because it gives the least total concrete once the band beam is counted.

Carbon and schedule

Item	Quantity	Factor	kg CO_2e/m^2
Concrete 5 ksi	7.29 joists + 0.93 band = 8.22 in = 0.209 m^3/m^2	365 kg/m^3	76.2
Rebar (joists 2.18 + beams 1.58 psf)	3.77 psf = 18.4 kg/m^2	854 kg/t	15.7
Total			**91.9**

Joist bar quantities use the CRSI-calibrated lengths (bottom 0.95L, top 0.60L per span) plus 0.0018bh temperature steel both ways in the slab. Schedule: pan joist cycle 7 days per 20,000 sf, +3.9% for rebar density (3.77 vs. 3.0 psf reference) = 7.3 days. Lead time 3 weeks.